Name:- Nidhi Manghnani
Roll No- 67
Batch T13
Subject- SEPM

THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE

Assignment-1

Explain Software process Models.

=> 1. A software process model is an abstract representation of a Software development process. It defines the specific flow of activities, actions and tasks required to build high quality Software. The primary goal is to to provide structure to what would otherwise be a chaotic development, environment ensuring that the final product meets requirements stays within budget and is delivered on time.

2) Fundamental process activities:
Every model, regardless of its structure, must address these four fundamental activities.

1. Software specification :- Defining the functionality and constraints of the software.
2. Software development : The actual design and programming of the system.
3. Software validation : Ensuring the software does what the customer wants.
4. Software Evolution: Modifying the software to adapt to changing user needs.

Waterfall Model:
waterfall Model is a famous and good version of SDLC (System Development lifecycle). For Software engineering. The waterfall model is a linear and sequential model, which means that a develop phase cannot begin until the previous phase is completed. We cannot overlap phases in waterfall model.

Key characteristics: Each phase must be complete and "signed off" befor the next begins.

phase : 1 : Requirement Analysis : Gathering all possible requirements of the system.

2. System Design: Designing the overall system architecture.

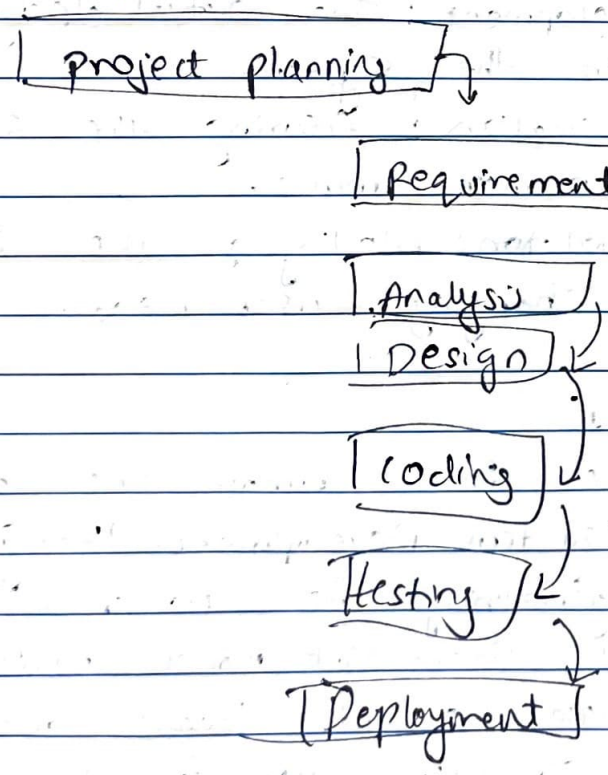
3. Implementation: Coding the system in small units.

4. Integration & testing : combining units and testing the whole system.

5. Deployment & maintainance: Releasing the product and fixing post release issues.

Advantages : Simple to understand ; easy to manage due to the rigidity of the model.

- Disadvantages : very difficult to go back and change requirements : no working software is produced until late in the life cycle

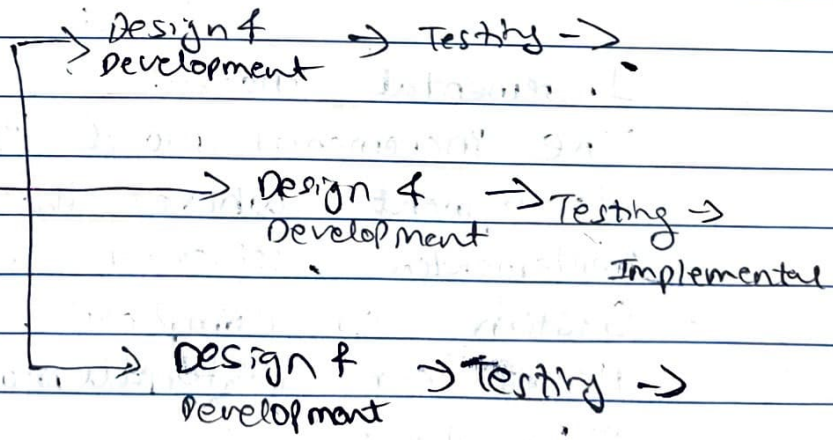


Spiral Model: The spiral model is an evolutionary

model that each "build" or increment must pass through these specific stages:

1. Requirement Analysis: The overall System requirements are defined, then specific features are assigned to a particular increment.
2. Design: The architecture of that specific module is planned to ensure it fits with the existing core.
3. Development: The actual programming of the module takes place.
4. Testing: The new increment is tested individually & then integrated with previous versions to check for compatability.
5. Deployment: The functional part is delivered to the customer for use & evaluation.

Advantages:
Users get working software quickly
Disadvantages:
High planning - it requires a clear "big picture" before starting.



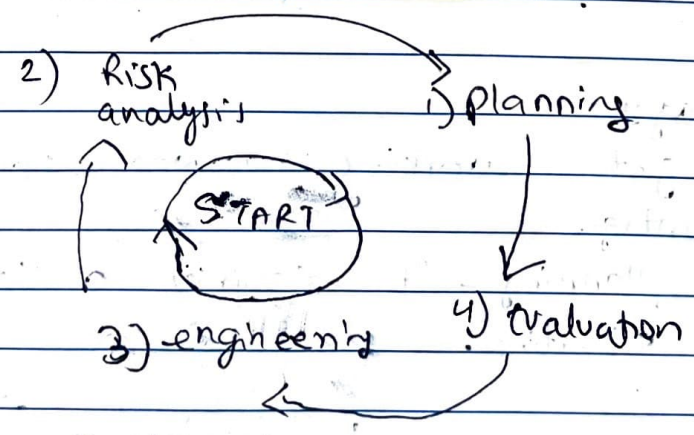
combines the itrative nature of prototyping with the controlled aspects of the waterfall model.

Key characteristics: It is divided into four quadrants. The item starts in the centre and moves in loops. Each loops involves:

1. Planning: Determining objectives and constraints.
2. Risk analysis: identifying and resolving technical management risks.
3. Engineering: Developing the next level of the product
4. Evaluation: Customers the progress before the next Spiral.

Advantages: Strongest model for risk management. Suitable for large, complex system.

Disadvantages: Can be very expensive, requires high expertise in risk assessment.

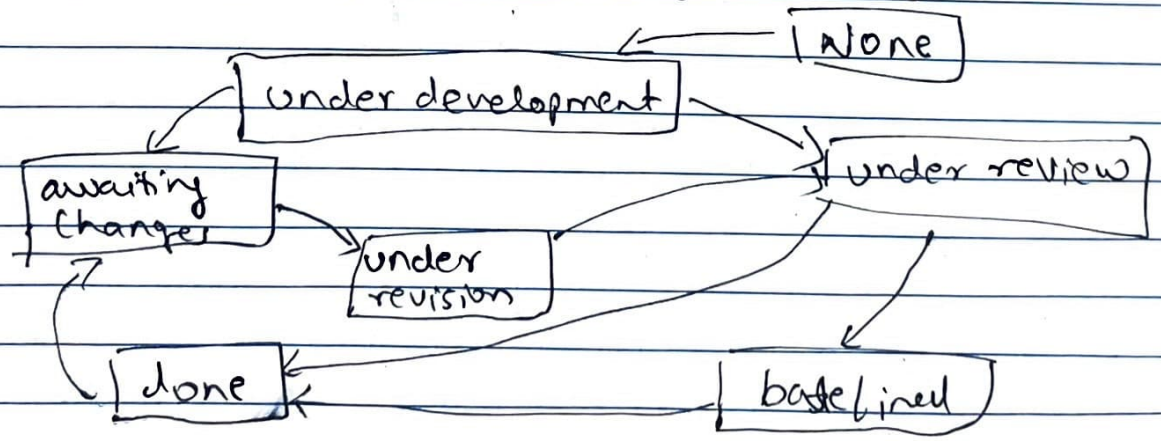


Incremental model:

The incremental model is a method of software development where the product is designed, implemented, delivered in pieces until the system is complete. It combines elements of the linear waterfall model with an iterative philosophy.

THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE

Concurrent model: The model is viewed as a network of activities. Each activity exists in a specific state at any given time. For example, the "Design" activity might be in the under development state, while the "Requirements" activity is in the Awaiting revision state.

In this model, "phases" are treated as concurrent activities, that communicate with each other.
Communication: Gathering requirements from the client.
Analysis: Modeling the data and functional requirements
Design: Creating the technical architecture.
Evolution: Coding and building the system.
Validation: continuous testing as components are completed.

Advantages: Significantly reduces the total time to market becomes tasks overlap.
Disadvantages: very difficult to manage, requires constant communication between teams.